

Year 7 · Chapter 1 · Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

- Column 1 example 1: $16 \rightarrow -16$
- Column 1 example 2: $7 \rightarrow -7$
- Column 2 example 1: $-(+26) \rightarrow$ = the opposite of 26; $= -26$
- Column 2 example 2: $-(-16) \rightarrow$ = the opposite of -16 ; $= 16$
- Column 3 example 1: The opposite of a number is -14 .
What is the number? $\rightarrow$ opposite of $?$ $= -14$; the number is 14
- Column 3 example 2: A diver is 7 m below sea level. Write this as an integer, then write its opposite. $\rightarrow$ below sea level: -7 ; opposite: 7

Questions (in order: Easy, Medium, Challenging)

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|-----------|-----------|--|
| 1. 59 | 15. 34 | 29. 29 |
| 2. -41 | 16. 2 | 30. 41 |
| 3. -9 | 17. 63 | 31. 50 |
| 4. -47 | 18. -8 | 32. 45 |
| 5. 25 | 19. -48 | 33. 22 |
| 6. -54 | 20. -50 | 34. 32 |
| 7. 14 | 21. 52 | 35. $-2, -1, 0, 1, 2$ |
| 8. 52 | 22. 4 | 36. 39 and -39 |
| 9. 43 | 23. -22 | 37. 26 |
| 10. -23 | 24. -9 | 38. $-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ |
| 11. -27 | 25. -48 | 39. 3 and -3 |
| 12. 21 | 26. -49 | 40. 6 |
| 13. 9 | 27. -57 | 41. $-3, 3$ |
| 14. -29 | 28. 25 | |

1.02 The number line (booklet p. 4)

WE DO examples (teacher)

- Column 1 example 1: 5 or $-4 \rightarrow 5$
- Column 1 example 2: -2 or $-3 \rightarrow -2$
- Column 2 example 1: -7 and 11 $\rightarrow -7$ to 0 is 7, 0 to 11 is 11; $7 + 11 = 18$
- Column 2 example 2: -8 and 3 $\rightarrow -8$ to 0 is 8, 0 to 3 is 3; $8 + 3 = 11$
- Column 3 example 1: How many integers are between -8 and 5? $\rightarrow -7$ to -1 : 7 integers, and 0; 1 to 4: 4 integers; $7 + 1 + 4 = 12$
- Column 3 example 2: How many integers are between -4 and 23? $\rightarrow -3$ to -1 : 3 integers, and 0; 1 to 22: 22 integers; $3 + 1 + 22 = 26$

Questions (in order: Easy, Medium, Challenging)

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|---------|----------|-----------|
| 1. 2 | 13. 15 | 25. 16 |
| 2. 8 | 14. 6 | 26. 20 |
| 3. 7 | 15. 0 | 27. -8 |
| 4. 1 | 16. 5 | 28. 3 |
| 5. 14 | 17. 6 | 29. -7 |
| 6. -2 | 18. -9 | 30. 12 |
| 7. 11 | 19. 10 | 31. 0 |
| 8. 14 | 20. 3 | 32. 55 |
| 9. -7 | 21. 1 | 33. -12 |
| 10. 3 | 22. 4 | 34. 42 |
| 11. 10 | 23. 23 | |
| 12. 3 | 24. 25 | |

1.03 Comparing and ordering integers

(booklet p. 5)

WE DO examples (teacher)

- Column 1 example 1: $-7 \square -8 \rightarrow >$
- Column 1 example 2: $-5 \square -10 \rightarrow >$
- Column 2 example 1: 15, $-8, -22, -9 \rightarrow$ smallest: -22 , largest: 15; $-22, -9, -8, 15$
- Column 2 example 2: 5, 14, $-24, -18 \rightarrow$ smallest: -24 , largest: 14; $-24, -18, 5, 14$
- Column 3 example 1: Write three integers greater than -14 but less than -9 . $\rightarrow$ integers from -13 to -10 ; e.g. $-13, -12, -11$
- Column 3 example 2: Which is the smallest: $-(-7)$, -7 or $-(-12)$? $\rightarrow -(-7) = 7, -(-12) = 12$; smallest: -7

Questions (in order: Easy, Medium, Challenging)

- | | | |
|---------|----------------------|--------------------------|
| 1. $>$ | 15. $<$ | 29. $-23, -14, 4, 15$ |
| 2. $<$ | 16. $>$ | 30. $-25, -16, 6, 20$ |
| 3. $>$ | 17. -7 | 31. 25, 12, $-27, -32$ |
| 4. $<$ | 18. -14 | 32. 9, $-13, -16, -22$ |
| 5. $>$ | 19. -14 | 33. $-2, -23, -31, -39$ |
| 6. $<$ | 20. -12 | 34. $-12, -9, 0, 2$ |
| 7. $>$ | 21. -21 | 35. e.g. $-17, -16, -15$ |
| 8. $>$ | 22. -26 | 36. -8 |
| 9. $>$ | 23. -24 | 37. $-7, -3, 0, 4$ |
| 10. $<$ | 24. -14 | 38. e.g. $-18, -17, -16$ |
| 11. $<$ | 25. -30 | 39. -6 |
| 12. $<$ | 26. $-20, 0, 3, 5$ | 40. $-11, -7, 0, 9$ |
| 13. $<$ | 27. $-24, -6, 5, 15$ | |
| 14. $>$ | 28. $-8, -7, 4, 10$ | |

1.04 Adding integers (booklet p. 6)

WE DO examples (teacher)

- Column 1 example 1: $-4 + 8 \rightarrow 4$
- Column 1 example 2: $-7 + 10 \rightarrow 3$
- Column 2 example 1: $-12 + (-11) + (-13) \rightarrow = -23 + (-13); = -36$
- Column 2 example 2: $-4 + (-3) + (-6) \rightarrow = -7 + (-6); = -13$
- Column 3 example 1: The temperature is -7°C and rises 8°C . What is the new temperature? $\rightarrow -7 + 8; = 1^\circ\text{C}$
- Column 3 example 2: A lift at level -3 goes up 6 floors. Which level is it on? $\rightarrow -3 + 6; =$ level 3

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|-----------|------------------------|
| 1. -6 | 15. 0 | 29. -10 |
| 2. 6 | 16. 6 | 30. -10 |
| 3. -1 | 17. 7 | 31. -2 |
| 4. -2 | 18. 10 | 32. 24 |
| 5. -9 | 19. -13 | 33. 7 |
| 6. -1 | 20. -1 | 34. 3 |
| 7. 2 | 21. -16 | 35. $-\$16$ |
| 8. -6 | 22. 7 | 36. 18 |
| 9. 1 | 23. 10 | 37. 10°C |
| 10. -4 | 24. 9 | 38. level 3 |
| 11. 0 | 25. 6 | 39. $-\$28$ |
| 12. 0 | 26. 5 | 40. 7 |
| 13. -3 | 27. -1 | |
| 14. -1 | 28. 17 | |

1.05 Subtracting integers (booklet p. 7)

WE DO examples (teacher)

- Column 1 example 1: $-6 - 4 \rightarrow -10$
- Column 1 example 2: $-5 - 4 \rightarrow -9$
- Column 2 example 1: $1 - (-15) + (-15) \rightarrow = 1 + 15 + (-15); = 16 + (-15); = 1$

4. Column 2 example 2: $7 - (-13) + (-6) \rightarrow = 7 + 13 + (-6); = 20 + (-6); = 14$
5. Column 3 example 1: The minimum temperature was -5°C and the maximum was 19°C . By how much did it rise? $\rightarrow 19 - (-5); = 19 + 5; = 24^{\circ}\text{C}$
6. Column 3 example 2: A submarine at -62 m rises 22 m . Where is it now? $\rightarrow -62 + 22; = -40\text{ m}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|---------|---------|--------------------------|
| 1. -15 | 14. -9 | 27. 11 |
| 2. -23 | 15. -10 | 28. 35 |
| 3. 0 | 16. -7 | 29. 15 |
| 4. -10 | 17. 21 | 30. 10 |
| 5. -8 | 18. 9 | 31. 9 |
| 6. -3 | 19. 11 | 32. -11 |
| 7. -4 | 20. 16 | 33. 30 m |
| 8. -2 | 21. 11 | 34. 21°C |
| 9. -6 | 22. -7 | 35. -83 m |
| 10. -8 | 23. 6 | 36. -13 |
| 11. -10 | 24. 21 | 37. 33 m |
| 12. 6 | 25. 4 | |
| 13. -18 | 26. -13 | |

1.06 Multiplying integers (booklet p. 8)

WE DO examples (teacher)

- Column 1 example 1: $-5 \times 3 \rightarrow -15$
- Column 1 example 2: $6 \times (-2) \rightarrow -12$
- Column 2 example 1: $4 \times (-5) \times (-3) \rightarrow = -20 \times (-3); = 60$
- Column 2 example 2: $1 \times (-1) \times 4 \rightarrow = -1 \times 4; = -4$
- Column 3 example 1: A diver goes down 2 m every minute. Where is she after 3 minutes? $\rightarrow -2 \times 3; = -6\text{ m}$
- Column 3 example 2: The temperature drops 4°C each hour for 3 hours. What is the total change? $\rightarrow -4 \times 3; = -12^{\circ}\text{C}$

Questions (in order: Easy, Medium, Challenging)

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|---------|----------|---------------------------|
| 1. 63 | 18. - | 35. 18 |
| 2. -49 | 19. + | 36. -100 |
| 3. 3 | 20. + | 37. -20 |
| 4. -3 | 21. - | 38. 16 |
| 5. 7 | 22. - | 39. 5 |
| 6. -72 | 23. + | 40. 24 |
| 7. -36 | 24. - | 41. -15 |
| 8. -64 | 25. - | 42. -8 |
| 9. -80 | 26. -36 | 43. 120 |
| 10. -5 | 27. 10 | 44. 3 |
| 11. 56 | 28. 150 | 45. -12 m |
| 12. -36 | 29. -1 | 46. -36°C |
| 13. 40 | 30. -60 | 47. -8 |
| 14. -56 | 31. -12 | 48. -28 m |
| 15. 30 | 32. -18 | 49. -45°C |
| 16. -24 | 33. -2 | |
| 17. - | 34. -100 | |

1.07 Dividing integers (booklet p. 9)

WE DO examples (teacher)

- Column 1 example 1: $36 \div (-4) \rightarrow -9$
- Column 1 example 2: $-28 \div (-7) \rightarrow 4$
- Column 2 example 1: $8 \div 4 \div 2 \rightarrow = 2 \div 2; = 1$
- Column 2 example 2: $-40 \div (-5) \div (-2) \rightarrow = 8 \div (-2); = -4$
- Column 3 example 1: A debt of $\$160$ is shared equally by 8 people. Write each share as an integer. $\rightarrow -160 \div 8; = -\20 each
- Column 3 example 2: The temperature fell 15°C over 3 hours. What was the change each hour? $\rightarrow -15 \div 3; = -5^{\circ}\text{C}$ each hour

Questions (in order: Easy, Medium, Challenging)

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|---------|--------|--------------------------|
| 1. -1 | 18. 9 | 35. -5 |
| 2. -10 | 19. 1 | 36. 1 |
| 3. 4 | 20. 5 | 37. -2 |
| 4. -5 | 21. -2 | 38. 2 |
| 5. -4 | 22. -1 | 39. 5 |
| 6. 8 | 23. 1 | 40. 3 |
| 7. -8 | 24. -1 | 41. -3 |
| 8. -8 | 25. -5 | 42. 2 |
| 9. 3 | 26. -5 | 43. 2 |
| 10. -4 | 27. -3 | 44. 32 |
| 11. 1 | 28. 2 | 45. $-\$19$ |
| 12. -10 | 29. -1 | 46. -2°C |
| 13. -4 | 30. 2 | 47. 20 |
| 14. -3 | 31. 1 | 48. $-\$18$ |
| 15. -5 | 32. -5 | 49. -6°C |
| 16. -1 | 33. -2 | |
| 17. -6 | 34. 3 | |

1.08 Order of operations with integers

(booklet p. 10)

WE DO examples (teacher)

- Column 1 example 1: $3 + (-6) \times (-6) \rightarrow 39$
- Column 1 example 2: $-9 - 5 \times (-6) \rightarrow 21$
- Column 2 example 1: $6 \times 1 + 2 \div 1 \rightarrow = 6 + 2; = 8$
- Column 2 example 2: $3 \times (-6) + (-16) \div 4 \rightarrow = -18 + (-4); = -22$
- Column 3 example 1: $7 - (-4 + 6) \times (-3) \rightarrow = 7 - 2 \times (-3); = 7 - (-6); = 13$
- Column 3 example 2: $6 - (-1 + (-2)) \times (-1) \rightarrow = 6 - (-3) \times (-1); = 6 - 3; = 3$

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|---------|
| 1. -29 | 15. -2 | 29. 1 |
| 2. -16 | 16. -27 | 30. 7 |
| 3. -36 | 17. -10 | 31. -26 |
| 4. 34 | 18. -21 | 32. -17 |
| 5. 22 | 19. -48 | 33. 14 |
| 6. 13 | 20. 9 | 34. -34 |
| 7. 19 | 21. -2 | 35. 34 |
| 8. 13 | 22. 13 | 36. 15 |
| 9. 19 | 23. -32 | 37. 34 |
| 10. 17 | 24. 5 | 38. 14 |
| 11. -22 | 25. 4 | 39. -14 |
| 12. -22 | 26. 29 | 40. -4 |
| 13. 8 | 27. -3 | |
| 14. 11 | 28. 1 | |

1.09 Integers on a calculator (booklet p. 11)

WE DO examples (teacher)

- Column 1 example 1: $252 + (-391) \rightarrow -139$
- Column 1 example 2: $-369 + 350 \rightarrow -19$
- Column 2 example 1: $(-86 - (-81)) \times (-16) \rightarrow$ bracket: $-5; -5 \times (-16) = 80$
- Column 2 example 2: $(-3 - 67) \times (-11) \rightarrow$ bracket: $-70; -70 \times (-11) = 770$
- Column 3 example 1: Estimate -60×44 , then calculate. $\rightarrow$ estimate: $-60 \times 40 \approx -2\ 400$; calculator: $-2\ 640$
- Column 3 example 2: Estimate -32×48 , then calculate. $\rightarrow$ estimate: $-30 \times 50 \approx -1\ 500$; calculator: $-1\ 536$

Questions (in order: Easy, Medium, Challenging)

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|---------|--------|----------|
| 1. 44 | 5. -14 | 9. -155 |
| 2. -284 | 6. -5 | 10. 91 |
| 3. 130 | 7. 413 | 11. -207 |
| 4. -121 | 8. 66 | 12. -330 |

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|------------|------------|----------------------------|
| 13. 308 | 24. -992 | 35. ≈ -600 ; -578 |
| 14. 324 | 25. -429 | 36. $\approx -2\ 400$; -2 |
| 15. 470 | 26. 1 716 | 088 |
| 16. 19 | 27. -190 | 37. $\approx -1\ 600$; -1 |
| 17. -172 | 28. -1 278 | 470 |
| 18. 66 | 29. 2 226 | 38. $\approx -1\ 500$; -1 |
| 19. -1 980 | 30. 140 | 288 |
| 20. -33 | 31. 2 831 | 39. $\approx -1\ 200$; -1 |
| 21. 900 | 32. 1 245 | 140 |
| 22. -264 | 33. -390 | 40. ≈ -600 ; -741 |
| 23. -561 | 34. -48 | 41. $\approx -1\ 500$; -1 |
| | | 248 |

1.10 Integer problems (booklet p. 12)

WE DO examples (teacher)

- Column 1 example 1: 5°C , rises $14^{\circ}\text{C} \rightarrow 19^{\circ}\text{C}$
- Column 1 example 2: 14°C , falls $4^{\circ}\text{C} \rightarrow 10^{\circ}\text{C}$
- Column 2 example 1: a hill 267 m above sea level and a reef 57 m below $\rightarrow 267 - (-57); = 324\text{ m}$
- Column 2 example 2: a hill 280 m above sea level and a reef 74 m below $\rightarrow 280 - (-74); = 354\text{ m}$

- Column 3 example 1: A diver is at -45 m . She rises 4 m a minute for 4 minutes. Where is she now? $\rightarrow 4 \times 4 = 16$; $-45 + 16; = -29\text{ m}$
- Column 3 example 2: A game scores $+4$ for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score? $\rightarrow 3 \times 4 = 12$; $5 \times (-3) = -15$; $12 + (-15) = -3$

Questions (in order: Easy, Medium, Challenging)

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|---------------------------|---------------------------|--------------------------|
| 1. -21°C | 13. -24°C | 25. \$55 |
| 2. -23°C | 14. 26°C | 26. 116 m |
| 3. 6°C | 15. 12°C | 27. 85 m |
| 4. 4°C | 16. -9°C | 28. 84 m |
| 5. 2°C | 17. \$20 | 29. 222 m |
| 6. 0°C | 18. $-\$112$ | 30. 445 m |
| 7. -6°C | 19. \$1 | 31. 174 m |
| 8. -17°C | 20. \$203 | 32. 116 m |
| 9. 21°C | 21. \$171 | 33. 7°C |
| 10. -1°C | 22. $-\$91$ | 34. -5 m |
| 11. -14°C | 23. \$25 | 35. 12 |
| 12. -4°C | 24. \$77 | 36. 27°C |